



LAB 9 Counter

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Technical Report

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Abstract: *FPGA Terasic
Board – ALTERA Cyclone IV EP4CE115F29C7N
IDE – Quartus Prime 18.1
Boolean Algebra
Digital Gates & Logical Comparators
Git & GitHub Control Version*

Keywords: *Counter, Sequential Logic, Flip-Flop, Asynchronous Reset Signal, Clock Cycle. & Preset.*

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1.- INTRODUCTION

For this laboratory, we were asked to create an 8-bit synchronous counter that, based on each clock cycle, would count the pulses, featuring an asynchronous reset to return the counter to zero, an enable input, and a preset that would reset the counter upon reaching a specific value. Through this, we put into practice the use of sequential logic elements, utilizing components to store bits of information and subsequently send that information to another block or follow-up logic. This is why it is called sequential — because it works with memory. This is important in digital systems since, combined with combinational logic, sequential logic supports the execution of more complex or robust processes, allowing for a broad range of applications. From this perspective, it greatly aids in defining processes within systems such as both volatile and non-volatile memories. It is crucial to understand all of this in detail to correctly implement all elements from various perspectives.

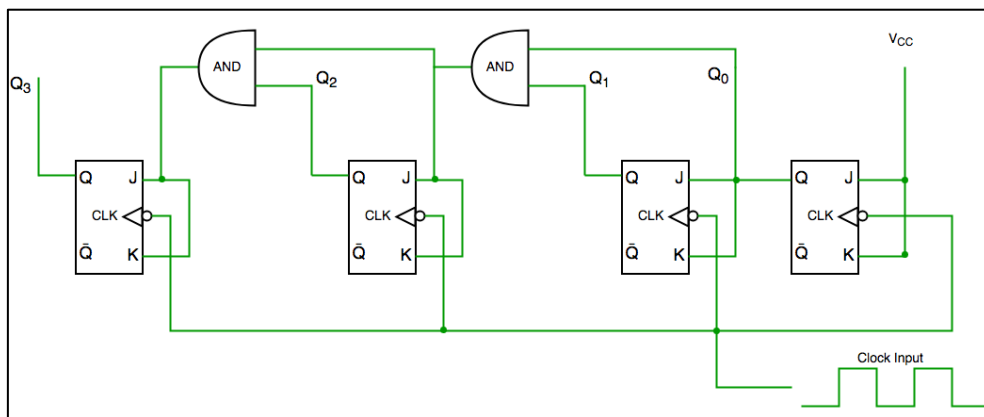


Figure: 1 Synchronous-Counter

1.1.- Requirements

Design an **8-bit synchronous** counter with the following characteristics:

- The counter must be 8 bits wide.
- The circuit must include an **asynchronous reset signal (arstn)**.
- On each clock cycle, the counter must **increase the output cnt [7:0] by 1**.
- The circuit must include an **enable signal (en)** to control counting.
- The counter must reset when the count reaches the value specified by the **preset [7:0] input**.

2.- FUNCTIONAL DESCRIPTION

For this laboratory, considerations were made to use certain megafunctions in order to improve the logic and organization of the circuit elements. Blocks were used for addition, comparison, a multiplexer, and the corresponding logic for D-type flip-flops, helping to maintain better organization. The main perspective of circuit is:

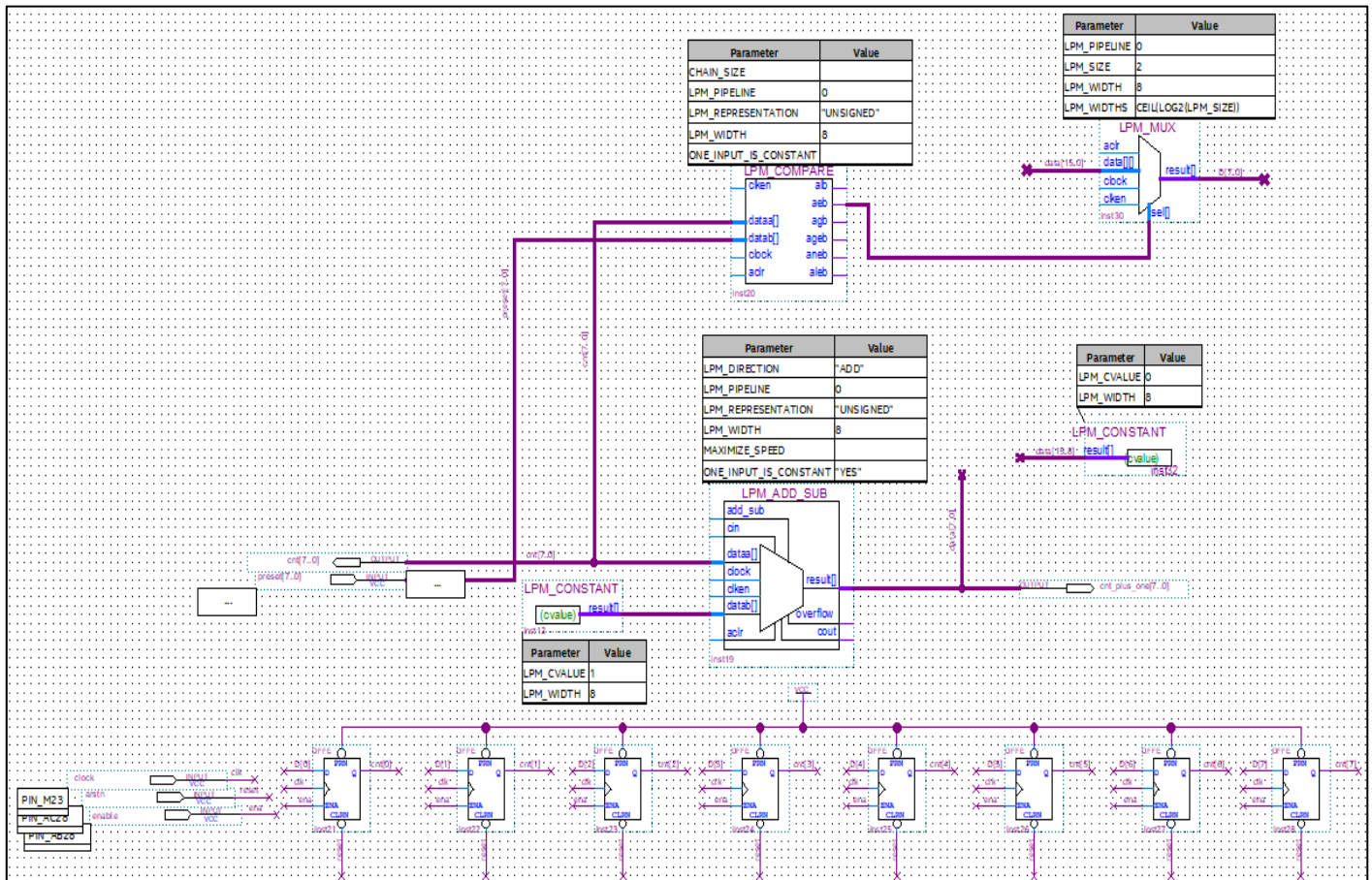


Figure: 2 ALU bdf

The main consideration was an array of eight flip-flops type D, with variant in Quartus “dffe” block:

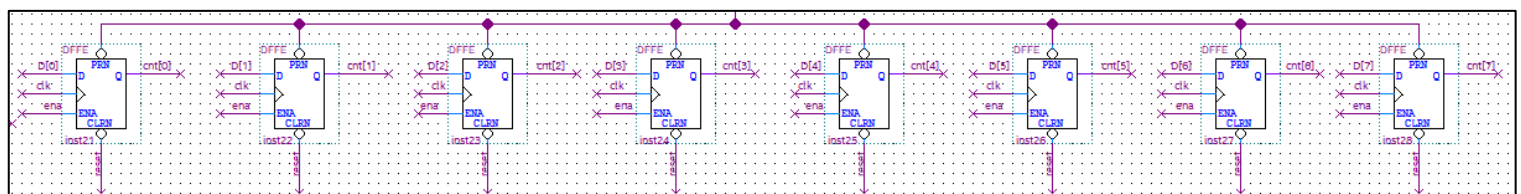


Figure: 3 Flip-Flop_D - Array

The pin assignment in the “Pin Planner” was as follows, in this case the consideration for clock was put in the “KEY0” because we need to see the value in each time of clock when we do a click in the button the counter add “1” in “1”.

Node Name	Direction	Location	I/O Bank
in arstn	Input	PIN_AC28	5
in clock	Input	PIN_M23	6
out cnt[7]	Output	PIN_G21	7
out cnt[6]	Output	PIN_G22	7
out cnt[5]	Output	PIN_G20	7
out cnt[4]	Output	PIN_H21	7
out cnt[3]	Output	PIN_E24	7
out cnt[2]	Output	PIN_E25	7
out cnt[1]	Output	PIN_E22	7
out cnt[0]	Output	PIN_E21	7
in enable	Input	PIN_AB28	5
in preset[7]	Input	PIN_Y23	5
in preset[6]	Input	PIN_Y24	5
in preset[5]	Input	PIN_AA22	5
in preset[4]	Input	PIN_AA23	5
in preset[3]	Input	PIN_AA24	5
in preset[2]	Input	PIN_AB23	5
in preset[1]	Input	PIN_AB24	5
in preset[0]	Input	PIN_AC24	5

Figure: 4 PinPlanner

RTL VIEWER

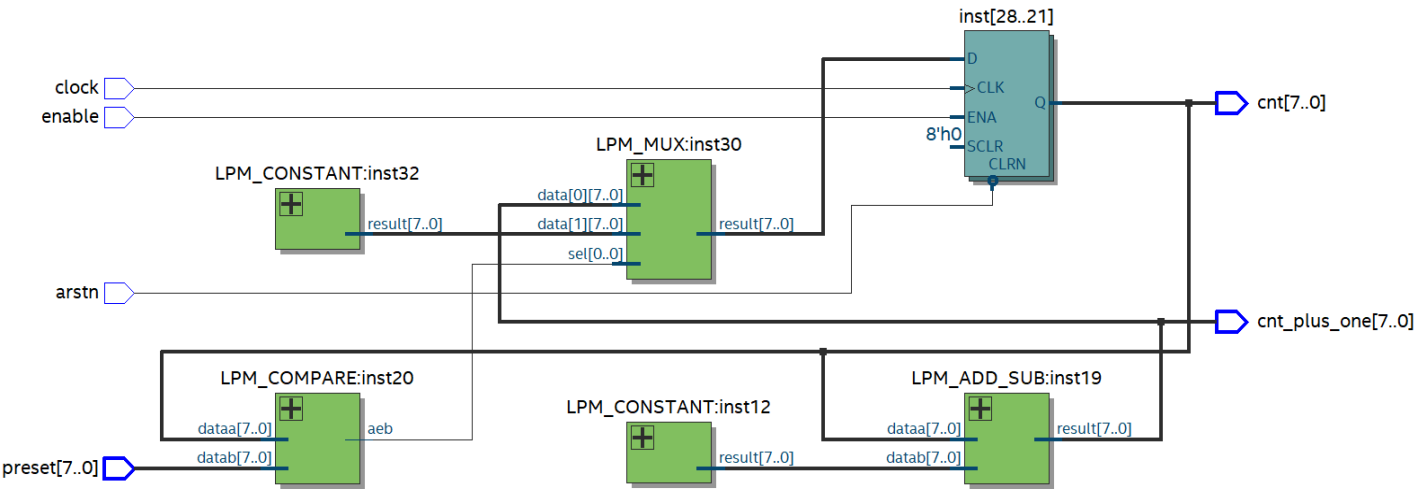


Figure: 5 RTL_Viewver

2.1.- Boolean Algebra Equations

For this stage was necessary set the transitions state equation:

1.- Model For Counter.

$$cnt(t+1) = \begin{pmatrix} 0, & \text{if } arstn = 0 \text{ (asynchronous reset).} \\ 0, & \text{if } (cnt(t) = preset) \wedge (en = 1) \text{ (asynchronous reset).} \\ cnt(t) + 1, & \text{if } (en = 1). \\ cnt(t) & \text{if } (en = 0). \end{pmatrix}$$

- Manual reset = to forcé "0".
- If cnt == preset and enable "1" counter reset to "0" automatically.
- If enable = 1, counter add +1 in each pulse of clock.
- If enable = 0, counter keep the value.

2.- Model For MUX.

$$D = \begin{pmatrix} 0, & \text{if } (cnt = preset). \\ cnt + 1 & \text{if } (cnt \neq preset). \end{pmatrix}$$

In hardware exist a main MUX that choose between 2 values.

- D = when cnt == preset.
- D = cnt +1 when the preset has not been reached.

Control for mux is:

$$select_mux = (cnt == preset)$$

3.- Flip-Flops (DFFE)- Equations.

$$Qi(t+1) = \begin{pmatrix} Di, & \text{if } (en = 1). \\ Qi(t), & \text{if } (en = 0). \end{pmatrix}$$

Where Di is the result from MUX (0 or cnt+1).

If arstn = 0:

$$Qi(t+1) = 0 \text{ (Asynchronous)}$$

2.2.- TrueTable & Results

The truth table for this circuit would be as follows: In this case we have in total 256 combinations.

arstn	en	preset	cnt[7:0] (actual)	cnt[7:0] (siguiente)
1	1	5	00000000	00000001
1	1	5	00000001	00000010
1	1	5	00000010	00000011
1	1	5	00000011	00000100
1	1	5	00000100	00000101
1	1	5	00000101	00000000
1	1	5	00000110	00000111
1	1	5	00000111	00001000
1	1	5	00001000	00001001
1	1	5	00001001	00001010
1	1	5	00001010	00001011
1	1	5	00001011	00001100
1	1	5	00001100	00001101
1	1	5	00001101	00001110
1	1	5	00001110	00001111
1	1	5	00001111	00010000
1	1	5	00010000	00010001
1	1	5	00010001	00010010
1	1	5	00010010	00010011
1	1	5	00010011	00010100

Figure: 6 TrueTable_Counter

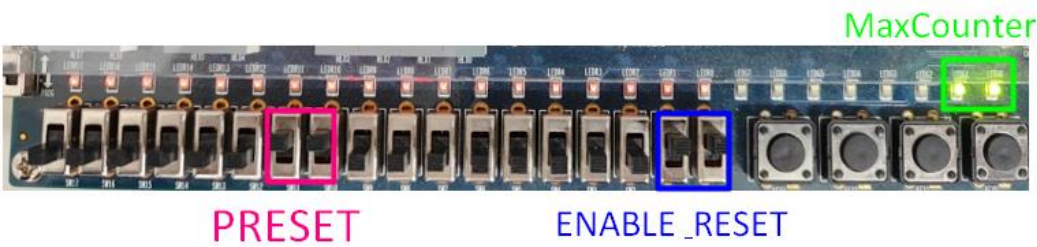
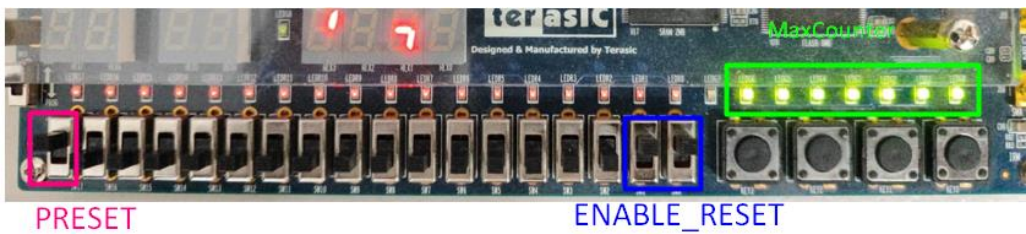


Figure: 7 Test1

ENABLE= 1

PRESET = 2

MaxCounter = Preset



ENABLE= 1

PRESET = 7

MaxCounter = Preset

2.3.- TrueTable & Results

A branch was created in the course repository corresponding to Lab9_Counter.

https://github.com/CeicGitHub/Fundamentals_Of_Digital_Design_FPGA_1/tree/QuartusLab9_Counter

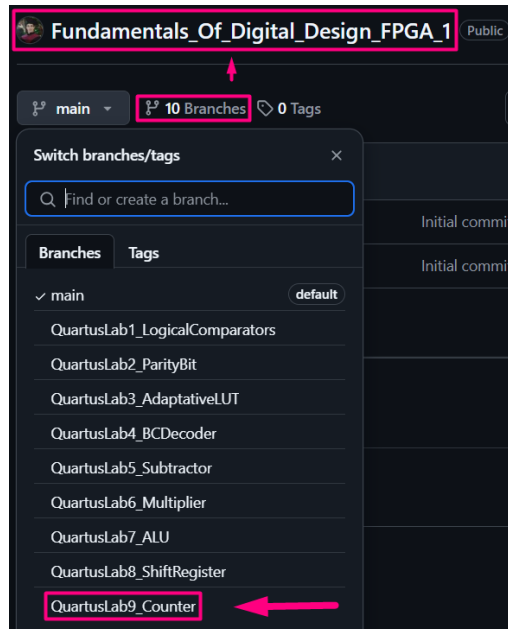


Figure: 8 GitHub Repository

3.- CHALLENGUES/ISSUES

Alonso Emmanuel López Macías

This laboratory exercise was somewhat challenging for us in terms of figuring out how to assign values to some of the Quartus library blocks. This was because previously the logic was implemented using primitive gates. For this part, we had to do some research on how to properly assign the correct values to ensure the program operated correctly.

Cesar Eduardo Inda Cenicerros

Some of the challenges of this practice involved understanding the logic of the counter, mainly how its behavior would operate through the functioning of the D-type flip-flop. A key part of this was determining what types of elements to use to replace the core principles of much of the circuit. This presented some difficulties when learning how to work with these functional blocks and how to parameterize the important data. Initially, part of the multiplexer represented synthesized functionalities, which greatly helped in understanding the circuit's overall functionality.

4.- CONCLUSIONS

Alonso Emmanuel López Macías

Personally, this laboratory allowed me to learn a great deal about the operation of a counter, from its functionality and logical implementation to the stage of digital design. I also gained a deeper understanding of how flip-flops work and the importance of grasping sequential logic, as well as how this hybrid format of processing — that is, combinational-sequential logic — can exist. Quartus software enables us to perform many functions that, in real life, would take a lot of time and resources to design conventionally. In a way, this digital representation serves as a synthesis of the integrated components we see today in many technological processes and devices, such as counters provided by companies like Texas Instruments or other electronics manufacturers around the world.

Cesar Eduardo Inda Cenicerros

In particular, this practice allowed me to realize the importance of understanding how sequential logic works, but mainly to grasp the underlying foundation behind the creation and operation of these digital elements. That is, starting from the basic principle of what a flip-flop is used for and the root reason for its existence in these systems, and how, together with other logic elements, it is possible to create a sort of memory that enables the recording of values through specific clock pulses. All these kinds of considerations intrigue me greatly because understanding this low-level operation allows one to comprehend all the higher layers, which becomes very useful for building a solid foundation in digital design fundamentals and knowing which elements to start from. I really enjoyed this practice because I was able to learn more about the elements available in the Quartus software.